

Question number	Answer	Additional guidance	Mark
9(a)(i)	A description that includes the following points: • use the (liver / rectal / core) temperature (immediately) (1) • use the graph and this temperature to get the {number of hours after death / time (after death)} (1) • subtract this value from the time of finding the corpse (1)	NB be careful not to award time of death ACCEPT work backwards using the graph	(3)

Question number	Answer	Additional guidance	Mark
9(a)(ii)	An explanation that includes the following points: • {other factors / named factor} affects {heat loss / rate of heat loss / body temperature (after death)} (1) • credit an explanation for how a named factor affects {heat loss / temperature} (1)	e.g. several hours since death body may have been moved to a cooler area e.g. {clothing / fat} {acts as an insulator / limits heat loss} cooler ambient temperature cools body faster	(2)

Question number	Answer	Additional guidance	Mark
9(b)(i)	not ethical / dead person cannot give their permission / relatives may not agree / humans are genetically diverse	IGNORE more available similar tissues	(1)

ion er	Answer	
ii)		
	Chemical tests	Discussion of usefulness
	pH	
	Pigs	Useful because decrease with time Useful as there is an increase with time with {small error bars / very little overlap in error bars} so timings are quite accurate
	Rats	Useful because decrease with time Useful as there is an increase with time but error bars overlap so would not give accurate timings
	Ammonia	
	Pigs	Useful as there is an increase with time Useful as there is an increase with time with no overlap in error bars so timings will be accurate
	Rats	Not very useful as {no clear patterns / no trend / data goes up and down} Not very useful because data goes up and down so one value will give two times Not useful as no ammonia after 48 hours Useful for determining that there was more than 48 hours since time of death as no ammonia present
	Hypoxanthine	
	Pigs	Useful as there is an increase with time Useful as there is an increase with time with no overlap in error bars so timings will be accurate Very low levels between 0 and 6 hours with overlapping error bars so may not be useful {then / until after 6 hours} There is an anomaly (at 24 hours) so this would need further investigation to see if this affects accuracy
	Rats	Not very useful as {no clear patterns / no trend / data goes up and down} Only useful up to {9 hours / 32 hours} Only useful up 6 hours otherwise one level could give two different timings
	Lactate	
	Pigs	Not very useful as {no clear patterns / no trend / data goes up and down} Could be useful up to 9 / 24 hours as there is an increase in levels / not useful after 9 / 24 hours as levels fluctuate
	Rats	Not very useful as {no clear patterns / no trend / data goes up and down} Could be useful up to 9 hours as there is an increase in levels / not useful after 9 hours as levels fluctuate
	General	small sample size so questionable significance use more than one marker and combine results results indicate that the test are species specific overall pigs are more useful than rats

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Level 1: Limited comments

1 mark = description of data for one chemical

2 marks = description of data for two chemicals

Level 2: Limited comments on usefulness

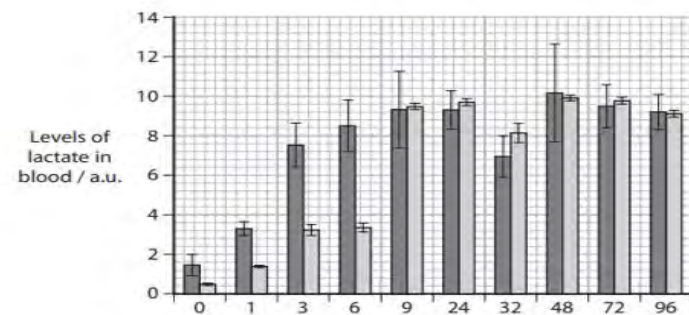
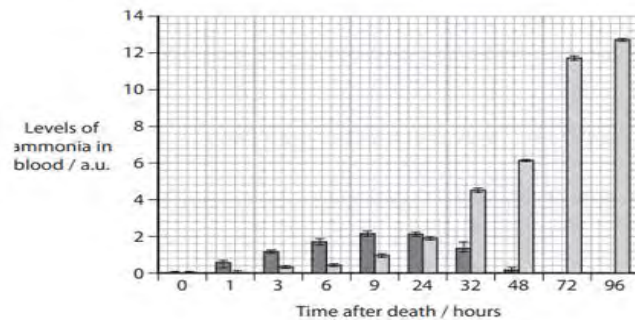
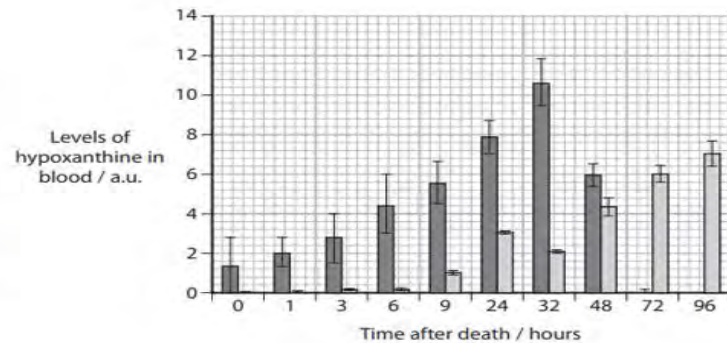
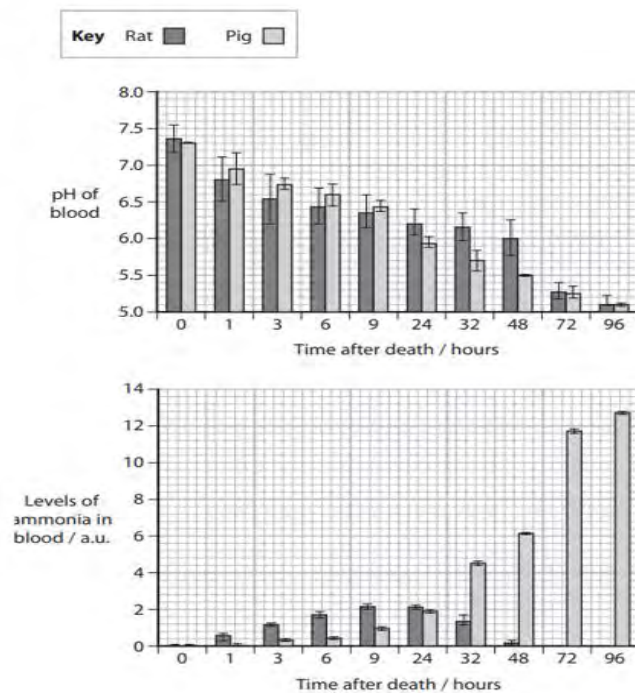
3 marks = usefulness of one chemical commented on OR one general comment

4 marks = usefulness of two chemicals commented on OR one chemical and one general comment

Level 3: Extended comments on usefulness

5 marks = usefulness of three chemicals commented on OR two chemicals and one general comment, that includes at least one extended comment

6 marks = usefulness four chemicals commented on OR three chemicals and one general comment, that includes at least one extended comment



Question number	Answer	Additional guidance	Mark
(b)(iii)	An answer that includes three of the following points: • {pH / some chemicals} seem to have similar patterns so could be useful (in humans) (1) • some patterns seem to be species-specific so may not be useful (1) • therefore testing will need to be done (on humans) / but will the testing be {allowed / possible} (1) • credit suitable comment about {sample size / error bars} (1) • chemical tests will need to be used in conjunction with other methods (1)	ACCEPT pH decreases in both species so could be useful (in humans) ACCEPT are different in pigs and rats humans may be similar to {pigs / rats}	(3)

